

Literature Status for arXiv:1804.01705 Question 6.2

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Status

Result type: literature_already_answered.

Verdict: Question 6.2 of arXiv:1804.01705 has an affirmative answer in later literature.

Original Question

The source paper is:

Rainis Haller, Jaan Kristjan Langemets, and Mart Poldvere, *Symmetric strong diameter two property*, arXiv:1804.01705.

In Section 6, Question 6.2, PDF page 16, the paper asks:

Does there exist a dual Banach space with the weak-star symmetric strong diameter two property and without the symmetric strong diameter two property?

Supporting Answer

The supporting paper is:

Rainis Haller, Jaan Kristjan Kaasik, and Andre Ostrak, *Separating diameter two properties from their weak-star counterparts in spaces of Lipschitz functions*, arXiv:2404.11430.

The supporting paper explicitly identifies the same question. In the introduction, PDF page 2, it states that it had remained open whether SSD2P and w^* -SSD2P are equivalent for dual spaces, citing [6, Question 6.2]. On PDF page 3 it says that its observation establishes that w^* -SSD2P differs from SSD2P in spaces of Lipschitz functions, thereby answering [6, Question 6.2].

The decisive statement appears in Section 3, PDF page 6: the authors state that they show there exist spaces of Lipschitz functions, hence dual Banach spaces, with w^* -SSD2P but without SSD2P, and that these are the first known examples answering [6, Question 6.2]. More concretely, for the metric space M from Haller–Oja–Poldvere Example 2.7, they use the known SLTP characterization to obtain w^* -SSD2P for $\text{Lip}_0(M)$, then prove that $\text{Lip}_0(M)$ does not have SSD2P.

Scope And Limitations

This is an already-known literature answer, not a new proof produced in this run. It clears exactly the 2018 question asking for a dual Banach space with w^* -SSD2P but without SSD2P.

It does not settle the separate question posed later in arXiv:2404.11430 asking whether there is a Banach space X with SSD2P such that X^{**} lacks SSD2P.

Run Evidence

The run's lightweight indexes were searched for arXiv ids 1804.01705 and 2404.11430 and for the key phrases w^* -SSD2P, SSD2P, and symmetric strong diameter two property. No existing durable packet for this exact source question was found.

References

References

- [1] R. Haller, J. K. Langemets, and M. Poldvere, *Symmetric strong diameter two property*, arXiv:1804.01705.
- [2] R. Haller, J. K. Kaasik, and A. Ostrak, *Separating diameter two properties from their weak-star counterparts in spaces of Lipschitz functions*, arXiv:2404.11430.